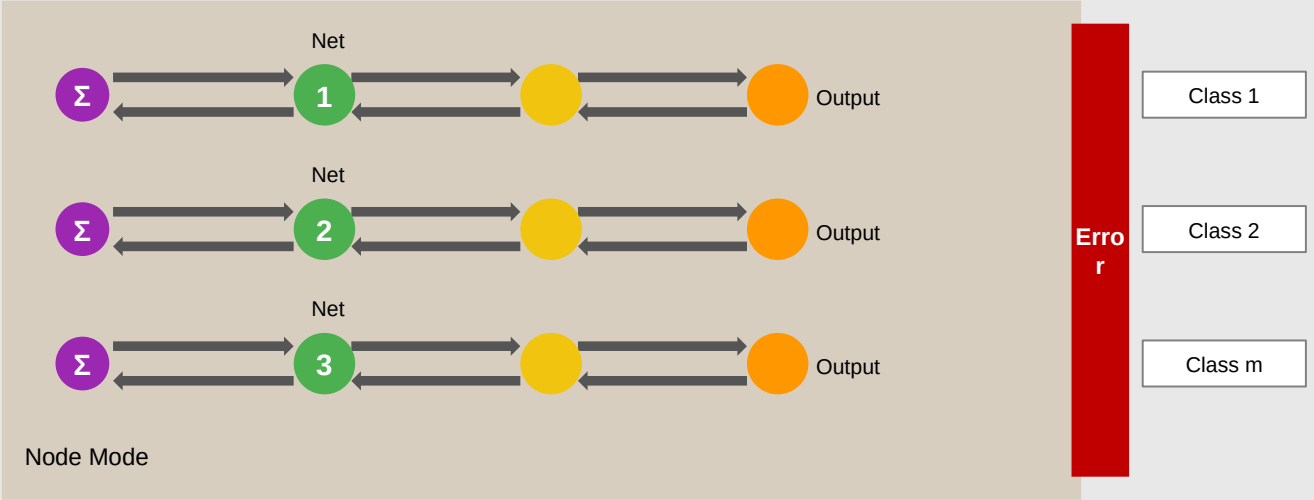
W_{ji} *Affects* Net_j $\Rightarrow$

Net_j *Affects* O_j $\Rightarrow$

O_j *Affects* $(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ $\Rightarrow$

$(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ *Affect Error*



$$\frac{\partial E}{\partial w_{ji}} = \frac{\partial E}{\partial Net_{k_1}}\frac{\partial Net_{k_1}}{\partial O_j} + \frac{\partial E}{\partial Net_{k_2}}\frac{\partial Net_{k_2}}{\partial O_j} + \ldots + \frac{\partial E}{\partial Net_{k_m}}\frac{\partial Net_{k_m}}{\partial O_j} \cdot \frac{\partial O_j}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial w_{ji}}$$

$$\frac{\partial E}{\partial w_{ji}} = \sum_{k=1}^{m} \frac{\partial E}{\partial Net_k}\frac{\partial Net_k}{\partial O_j} \cdot \frac{\partial O_j}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial w_{ji}} = \sum_{k=1}^{m} -\delta_k w_{kj} \cdot O_j(1-O_j) \cdot O_i$$

$$\Delta_{ji} = -\eta\frac{\partial E}{\partial w_{ji}} = \sum_{k=1}^{m} \delta_k w_{kj} \cdot O_j(1-O_j) \cdot O_i = \eta\delta_j \cdot O_i$$

Where $\delta_j = \sum_{k=1}^{m} \delta_k w_{kj} \cdot O_j(1-O_j)$ (for each Node j at hidden layer)

$$\frac{\partial E}{\partial Net_k} \rightarrow -\delta_k$$

$$Net_k = \sum_j w_{kj} O_j \rightarrow \frac{\partial Net_k}{\partial O_j} = w_{kj}$$

$$O_j = \frac{1}{1+e^{-Net_j}} \rightarrow \frac{\partial O_j}{\partial Net_j} = O_j(1-O_j)$$

$$Net_j = \sum_i w_{ji} O_i \rightarrow \frac{\partial Net_j}{\partial w_{ji}} = O_i$$